

POCKETSMART AI: BRAINSTORMING & IDEA PRIORITIZATION

1. Overview & Brainstorming Goals

The primary objective of the ideation phase was to design an intelligent, accessible, and automated personal financial planning ecosystem. Traditional budgeting tools rely on static spreadsheets and manual calculations without considering category-specific constraints. The ideation focus was centered on how Large Language Models (LLMs) and heuristic algorithms can dynamically solve unstructured domestic expenditure problems.

2. Candidate Feature Generation (Brainstormed Ideas)

During the collaborative ideation phase, multiple innovative concepts were proposed:

- **Idea 1: Multi-Category Modular Planners:** Segregating complex budgeting into distinct lifestyle categories (Home Interior, Party/Event, and Jewelry/Fashion) rather than a generic monthly expense tracker.
- **Idea 2: Automated E-Commerce Affiliate Search Routing:** Programmatically parsing recommended items into search queries with direct buy routes across Indian e-commerce platforms (Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, Tanishq, Myntra).
- **Idea 3: Multimodal Visual Processing:** Enabling users to upload reference images (e.g., jewelry ornaments or room layouts) to extract aesthetic styles and match budget parameters.
- **Idea 4: Fail-Safe High-Availability Engine:** Architecting a resilient backend layer that calculates weighted percentage

budgets deterministically if external AI APIs experience latency or quotas.

- **Idea 5: Microservices vs. Decoupled Monolith:** Utilizing FastAPI's asynchronous ASGI structure with decoupled routers and JWT security for low-latency client performance.

3. Prioritization Framework (Impact vs. Feasibility Matrix)

To optimize development within academic constraints, brainstormed features were evaluated using a 2x2 Impact-Feasibility Matrix and MoSCoW categorization:

Proposed Feature	Business Impact	Technical Feasibility	Priority Category	Implementation Status
Home Interior Budget Planner	High	High	Must Have (P0)	Fully Implemented
Party & Event Planner	High	High	Must Have (P0)	Fully Implemented
Multimodal Jewelry Vision Route	High	Medium	Must Have (P0)	Fully Implemented
JWT Session Authentication	High	High	Must Have (P0)	Fully Implemented
E-Commerce Search Deep-Linking	High	High	Should Have (P1)	Fully Implemented
Deterministic Fallback Engine	Critical	High	Must Have (P0)	Fully Implemented
Direct Bank API Synchronization	Medium	Very Low	Won't Have (P3)	Deferred to Future Scope

Multi-Currency Live FX Converter	Low	Medium	Could Have (P2)	Deferred to Future Scope
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4. Final Selected Solution Strategy

Based on the prioritization matrix, **PocketSmart AI** was formulated as a modular web application powered by Python 3.12, FastAPI, Bootstrap 5, and Google Gemini. The system maximizes user impact by automating cost trade-offs and delivering instant shopping links within sub-second response times.